

When Optimization Outruns Feedback

The Feedback Horizon Gap in AI Optimization

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Abstract

AI systems increasingly improve by running fast generate–evaluate–update loops against formal verifiers, tests, reward models, benchmarks, simulations, or other evaluators. Some of these evaluators are close to the target of interest. Others are proxies whose important errors become visible only later. The word “feedback” hides a count: how many proxy-guided optimization updates occur before sufficiently independent target-level evidence becomes available. Call that count K . When updates are approximately periodic, $K \approx \tau_T/\tau_P$, where τ_P is the proxy-feedback latency and τ_T the target-evidence latency. Target loss decomposes exactly as

$$L_T(\theta_K) - L_T(\theta_0) = \sum_{i=0}^{K-1} [L_T(\theta_{i+1}) - L_T(\theta_i)].$$

Systematic proxy error can therefore accumulate for K selected updates before independent evidence can correct the process. Reward-model overoptimization shows the failure mode in AI. Formal theorem proving and executable code show the opposite case, where rapid verification stays tied to the target. Weather forecasting, autonomous laboratories, and surrogate clinical endpoints sit in between: optimization and target evidence run on different clocks. As optimization accelerates, the operative variable is how many consequential updates occur between independent target checks, not merely how quickly an evaluator returns a score.

1 Introduction

Problems with clear, cheap verification are unusually friendly to optimization. A compiler rejects invalid syntax. A unit test can reject a broken implementation. Lean rejects an invalid formal proof. A game engine returns an exact win or loss. In each case, candidate generation can be coupled to a short feedback loop whose output is strongly connected to the local target.

Many important problems do not have this structure. A reward model can score a response immediately while human preference outside the reward model’s training distribution is only sampled occasionally. A weather model can produce a ten-day forecast in under a minute while the ten-day atmosphere still takes ten days to occur (Lam et al. 2023). A drug may improve a surrogate endpoint months or years before its effect on survival or quality of life is known, and confirmatory evidence can arrive years after an accelerated approval (Gyawali, Hey, and Kesselheim 2019; Liu, Kesselheim, and Cliff 2024). In these cases, “feedback speed” is not one number.

Optimization runs on one loop; independent target evidence arrives on another. The gap is measured in *selected optimization updates* between independent target checks, not wall-clock latency alone.

Let K denote that count. If proxy updates occur at roughly regular intervals τ_P and informative target evidence arrives every τ_T , then

$$K \approx \left\lfloor \frac{\tau_T}{\tau_P} \right\rfloor. \tag{1}$$

Parallel search, batching, asynchronous evaluation, and variable update sizes can make wall-clock time a poor proxy. Count updates directly.

Large K can coexist with safety. When the proxy evaluator matches the target criterion, millions of updates may do no harm. AlphaProof uses Lean-verified outcomes as grounded feedback and learns from millions of formally checked problems (Hubert et al. 2025). The failure mode appears when repeated selection acts on an imperfect proxy whose errors stay hidden across many updates before an independent check.

Three questions often get conflated:

- (i) How fast can candidates be generated?
- (ii) How fast can the signal used for optimization be computed?
- (iii) How fast can sufficiently independent evidence about the target become available?

AI can compress the first two by orders of magnitude without necessarily compressing the third.

2 Timescales and Accumulation

2.1 Four timescales

We distinguish four latencies:

$$\tau_G = \text{candidate-generation latency}, \quad (2)$$

$$\tau_P = \text{latency of feedback actually used for optimization}, \quad (3)$$

$$\tau_T = \text{latency until informative target-level evidence is available}, \quad (4)$$

$$\tau_C = \text{latency until that evidence can materially correct the optimizer or system}. \quad (5)$$

The target need not be metaphysical “ground truth.” It is the outcome we actually care about for the analysis. The key requirement is that target evidence have at least partially independent failure modes from the proxy used to select updates. For a reward model, a fresh human evaluation can serve as target evidence. For a ten-day weather forecast, the realized atmosphere does. For an accelerated drug approval based on tumor response, overall survival or quality of life can provide a more target-relevant endpoint.

Usually, but not always,

$$\tau_G \leq \tau_P \leq \tau_T \leq \tau_C.$$

A slow proxy evaluator can have $\tau_P > \tau_T$, and target evidence may exist before an institution notices or acts on it.

2.2 The feedback horizon

Let θ_i be the system state after the i -th proxy-guided update following an independent target check. Define

$$K = \text{number of proxy-guided updates before the next independent target check}. \quad (6)$$

Equation (1) follows only for approximately regular sequential updates.

The distinction matters for AI systems because a million candidates may be generated and filtered in parallel during one hour. Wall-clock latency says “one hour.” The relevant exposure can be closer to one million selected comparisons or update opportunities.

2.3 Accumulation identity

Let $L_T(\theta)$ denote target loss, with lower values better. Define the target effect of proxy-guided update i as

$$\Delta_i = L_T(\theta_{i+1}) - L_T(\theta_i).$$

Then exactly,

$$L_T(\theta_K) - L_T(\theta_0) = \sum_{i=0}^{K-1} \Delta_i. \quad (7)$$

The identity holds regardless of the optimizer, the proxy, linear response, smoothness, stationarity, or whether the target is observable during optimization.

Equation (7) separates how long error can accumulate from whether it does. A perfect evaluator yields $\Delta_i \leq 0$ under selection; a large K then helps. An exploitable proxy yields positive Δ_i that can stack for many selected updates before independent evidence arrives.

Under the additional assumption that Δ_i are identically distributed with mean μ ,

$$\mathbb{E}[L_T(\theta_K) - L_T(\theta_0)] = K\mu. \quad (8)$$

Systematic per-update target drift then accumulates linearly in the number of unchecked updates.

If $\mu = 0$, $\text{Var}(\Delta_i) = \sigma^2$, and the errors are independent,

$$\text{SD}\left(\sum_i \Delta_i\right) = \sigma\sqrt{K}. \quad (9)$$

More generally, for weakly stationary update effects with autocovariance $\gamma_h = \text{Cov}(\Delta_i, \Delta_{i+h})$,

$$\text{Var}\left(\sum_{i=0}^{K-1} \Delta_i\right) = K\sigma^2 + 2\sum_{h=1}^{K-1} (K-h)\gamma_h. \quad (10)$$

Persistent positive correlation in proxy errors therefore makes repeated optimization more dangerous than independent noise. This is the regime one would expect when an optimizer repeatedly exploits the same evaluator blind spot.

2.4 Reading K in context

A theorem checker and a learned reward model can share the same K and differ sharply in reliability. Interpretation turns on:

- **Fidelity:** how informative is proxy improvement about target improvement?
- **Independence:** are later target checks genuinely independent, or do they reuse the same blind spots?
- **Reversibility:** can proxy-driven changes be cheaply undone?
- **Decomposability:** can target failure be detected locally before the full target horizon?
- **Consequence per update:** how far can one accepted update move the deployed system?

3 Feedback Regimes

The distinction between proxy and target feedback becomes more useful when combined with the *kind* of feedback available. A short feedback horizon is not equally valuable in every regime. What matters is both how rapidly evidence arrives and what that evidence establishes.

We distinguish six broad regimes. They are not mutually exclusive. A single system can move through several of them as an artifact progresses from simulation to deployment.

Exact symbolic feedback. A candidate is checked against a formally specified criterion, and acceptance or rejection follows mechanically. Formal theorem proving is the clearest example. Once a mathematical statement has been faithfully represented in Lean, a proof checker provides an unusually tight relationship between the feedback used during search and the local target of formal validity (Hubert et al. 2025). Rule-complete games have a similar structure at the level of terminal outcomes: the rules determine whether a position is legal and who won (Silver, Schrittwieser, Simonyan, et al. 2017).

This regime can support extremely large search because verification is cheap relative to generation. Its main residual failure lies outside the verifier: the formal statement, game objective, or specification may fail to represent what was intended. Thus exact verification solves

“Does x satisfy specification S ?”

without solving

“Is S the right specification?”.

Executable feedback. The candidate is run, and behavior is compared with tests, assertions, constraints, or measured performance. Compilation, unit tests, property tests, exploit reproduction, and hardware benchmarks belong here. Execution feedback enables LLMs to debug generated programs iteratively (Chen et al. 2023). Algorithm optimization can similarly combine cheap correctness checks with less frequent direct measurements of runtime (Mankowitz, Michi, Zitnick, et al. 2023).

Executable feedback differs from exact symbolic feedback because the tested behavior is normally only a sample of possible behavior. A program can pass all available tests while remaining insecure, brittle, or incorrect outside the tested distribution. The feedback horizon is therefore property-specific:

$$\tau_T^{\text{tested property}} \ll \tau_T^{\text{unobserved property}}.$$

There is no single feedback horizon for “the program.”

Repeated empirical feedback. The system acts on the world and obtains direct measurements sufficiently quickly to guide the next action. Closed-loop chemistry and autonomous materials laboratories are increasingly close to this regime. AlphaFlow uses in-situ measurements to guide subsequent experiments (Volk et al. 2023), while A-Lab integrates experiment selection, robotic execution, synthesis, and characterization in one autonomous loop (Szymanski et al. 2023).

Here the verifier is not purely computational. Nature performs part of the computation. The cycle therefore has a physical floor set by manipulation, reaction, growth, fabrication, measurement, or other dynamics of the system being studied. Automation can remove human handoffs, but it cannot in general make $\tau_T \rightarrow 0$.

Noisy statistical feedback. Many important outcomes are not determined by one observation. Evidence accumulates over repeated measurements, noisy labels, heterogeneous cases, or stochastic outcomes. Preference learning and clinical trials are examples. A reward model compresses many human judgments into a rapidly queried statistical evaluator; clinical surrogate endpoints compress later patient outcomes into earlier measurements.

In this regime, a feedback event need not be wrong in isolation. The problem is that its evidential relationship to the target is probabilistic:

$$P(T \mid F) \neq 0, 1.$$

Repeated optimization can therefore exploit stable residual error in the mapping from F to T . Reward-model overoptimization demonstrates this directly: independently evaluated quality can fall while the learned reward continues to rise (Gao, Schulman, and Hilton 2022; Rafailov et al. 2024). Clinical surrogate endpoints show the same structural distinction outside AI: an earlier endpoint can be useful while remaining an imperfect predictor of survival or quality of life (Gyawali, Hey, and Kesselheim 2019; Naci, Smalley, and Kesselheim 2017; Liu, Kesselheim, and Cliff 2024).

Sparse historical feedback. Some targets are observed only after long delays or rare events. Examples include major accidents, long-horizon institutional outcomes, macroeconomic policy effects, and sufficiently rare safety failures. In such settings it may be impossible to generate repeated target-level observations on demand.

The main difficulty is therefore not merely noisy evaluation but low target-observation rate. If catastrophic failures occur once per 10^6 system-hours, a million hours of safe operation is still only weak evidence about failure probabilities much below that scale. Mature safety fields respond by constructing intermediate evidence: near misses, precursor events, stress tests, component reliability estimates, simulation, and probabilistic risk models. These do not eliminate the long target horizon. They attempt to infer the target from more frequent observations.

Endogenous or reflexive feedback. Finally, the system being optimized may influence the process that evaluates it. Recommendation algorithms affect the users whose clicks constitute future training data. Economic policy changes the economy from which subsequent measurements are taken. A strategically capable AI may alter, anticipate, select for, or otherwise condition on its evaluators.

The distinction is structural. In the earlier regimes we can approximately write

$$\theta \longrightarrow \text{outcome} \longrightarrow F,$$

with F treated as an external observation. In a reflexive regime,

$$\theta \longrightarrow \{\text{outcome, measurement process, future data}\} \longrightarrow F.$$

The evaluator is now part of the environment being optimized.

This is more demanding than ordinary proxy error. Even an initially high-fidelity evaluator can become less informative under optimization because the distribution or the measurement process changes. Real-time engagement optimization provides a mild example: clicks can be measured almost instantly, yet optimizing them need not optimize the user’s ideal preferences or welfare (Khambatta et al. 2023). More adversarial forms are a central concern in AI alignment, but are not yet supported by a comparably clean empirical literature.

The feedback-horizon quantity K is most informative when read together with this classification. Large K under exact symbolic feedback can be benign. Moderate K under a systematically exploitable statistical evaluator can already be problematic. Under sparse or endogenous feedback, even defining an independent target check may be the hard part.

4 Direct Evidence: Reward-Model Overoptimization

Reward-model optimization instantiates Equation (7). In standard RLHF, a learned reward model predicts human preferences and is then optimized by a policy. The reward model is cheaper to query than humans and therefore becomes the fast proxy loop.

Gao, Schulman, and Hilton constructed a controlled setting in which a fixed “gold” reward model supplied labels used to train a weaker proxy reward model (Gao, Schulman, and Hilton 2022). They then optimized the policy against the proxy using reinforcement learning or best-of- n sampling. As proxy optimization increased, gold-model performance eventually degraded. What matters empirically is the separation:

$$\text{proxy score} \uparrow \quad \text{while} \quad \text{independent target proxy} \downarrow .$$

The optimizer had enough search power to discover outputs that scored well according to the learned evaluator but worse according to the independently held reference evaluator.

Rafailov et al. extended the observation to direct alignment algorithms such as DPO, which do not expose a separate learned reward model during optimization (Rafailov et al. 2024). They found analogous deterioration at higher optimization budgets and reported that degradation could occur before one full epoch of the preference dataset was completed. Overoptimization is not confined to explicitly training against a separate scalar reward network. Repeated selection against a finite preference signal can move the policy into regions where the optimization signal ceases to track the broader evaluation criterion.

More recent work interprets the same phenomenon partly as distribution shift. Ackermann, Ishida, and Sugiyama show that as the optimized policy moves away from the reward model’s training distribution, the reward-model parameter estimate and resulting policy gradient can become inconsistent; an off-policy correction improves final policy quality in their experiments (Ackermann, Ishida, and Sugiyama 2025). Iterated RLHF can reduce overoptimization when new reward models better approximate the target distribution, but early overoptimization is not always fully recovered (Wolf, Kirk, and Musolesi 2025).

In the present notation, reward-model training creates a fast evaluator P . Policy optimization can perform many updates before fresh human labels or otherwise independent target checks update P . Wall-clock τ_T varies across RLHF pipelines. The qualitative pattern is stable: continued optimization against a fixed imperfect evaluator increases accumulated target error before the evaluator is corrected.

Goodhart’s law names proxy–target divergence. The feedback horizon counts how much *selection* can run before that divergence meets an independent check.

5 Large K with Strong Verifiers

Formal theorem proving and executable software carry large K without the reward-model failure mode.

5.1 Formal theorem proving

Formal proof assistants sharply separate candidate generation from verification. Lean can reject an invalid tactic or proof according to a fixed formal semantics. AlphaProof turns this into an RL environment: tactics are actions, Lean states are observations, and Lean-verified proof, disproof, or timeout outcomes provide grounded feedback (Hubert et al. 2025). Its training uses millions of auto-formalized problems, and its test-time RL can generate and learn from millions of related variants. At the 2024 IMO, the AlphaProof-centered system solved three non-geometry problems, with difficult solutions requiring two to three days of test-time RL (Hubert et al. 2025).

This is a large- K regime by construction. The system makes enormous numbers of search and learning moves before any human reads the final proof. The local target, formal validity in Lean, is checked inside the fast loop.

Formal verification checks the formal statement. If the formalization misses the intended natural-language problem, the verifier can be perfectly reliable about the wrong target. AlphaProof therefore splits into two clocks:

proof validity: $\tau_P \approx \tau_T$ once formalized,
faithful formalization: τ_T may require separate human or semantic checking.

5.2 Executable software

Execution feedback gives code generation a similarly short local loop. Chen et al. showed that LLM self-debugging using execution results and feedback messages improves code-generation performance by up to 12 percentage points on benchmarks with unit tests and can match or exceed baselines generating over ten times as many candidate programs (Chen et al. 2023). Here the test harness supplies dense, actionable information about some classes of failure.

Human development benefits from the same structure. In a randomized experiment with 95 professional developers implementing the same JavaScript HTTP server, developers using GitHub Copilot completed the task in an average of 1 hour 11 minutes versus 2 hours 41 minutes without Copilot, a 55.8% reduction in completion time (Peng et al. 2023). The submissions were evaluated with a test suite.

Software often mixes both regimes in one artifact:

fast trusted feedback for covered properties + slow or missing feedback for uncovered properties.

The same codebase can pass unit tests quickly while security and maintainability failures surface only under deployment or audit (Fu et al. 2023).

6 Matched Feedback Horizons

Table 1 compares fast proxy loops, current AI cycles, human or classical cycles, target-evidence latency τ_T , and approximate K across sixteen domains. Bracket tags mark provenance: [M] directly measured or reported in the cited source; [D] simple derivation from reported quantities; [E] order-of-magnitude or workflow estimate; [NQ] not defensibly quantified from available literature; [S] structural analogy rather than direct empirical evidence.

Table 1: Matched feedback structures across AI and human systems. τ_G is candidate-generation latency, τ_P the latency of feedback actually used for selection or updating, and τ_T the latency until sufficiently independent evidence about the target is available. K is reported only where the notion of repeated proxy-guided updates before a target check is meaningful. Platform throughput is not silently converted into serial latency. Domain-by-domain readings appear in Subsection 6.1.

Domain	Fast loop / proxy	Current AI cycle	Human / classical cycle	Target-level evidence τ_T	evi- Limiting cycle	Approx. K / status
Formal theorem proving	Formal proof checker, e.g. Lean acceptance/rejection.	Lean verification is inside the search loop. AlphaProof used millions of formally checked problem variants; difficult IMO solutions required 2–3 days of test-time RL end-to-end [M] (Hubert et al. 2025).	Human proof construction and review are typically hours to days for nontrivial problems [E]; no matched cycle-time study is available.	For the target “valid proof of this formal statement”, target evidence arrives when the proof checker accepts or rejects.	Proof-checking computation.	Potentially enormous K of search steps, but effectively $K \simeq 1$ relative to formal validity.
Executable software	Compilation, execution, unit tests, property tests.	LLMs can repeatedly execute and self-debug generated programs; per-iteration latency is implementation-dependent [NQ] (Chen et al. 2023).	In a randomized coding experiment, the control group required 2 h 41 min on average versus 1 h 11 min with Copilot [M] (Peng et al. 2023).	Seconds or minutes for covered functional properties [E], but security, maintainability, performance under real workload, and rare bugs can remain latent much longer.	Execution plus test runtime for covered properties.	Property-dependent.
Rule-complete games	Exact game rules and terminal win/loss.	AlphaGo Zero learned entirely from self-play; every completed game provides an exact terminal result [M] (Silver, Schrittwieser, Simonyan, et al. 2017).	Humans normally receive a correct game result after each played game; wall-clock duration depends on the game and time control [NQ].	The task-level target, winning under the stated rules, is observed at game termination.	Simulator/game execution.	Approximately one terminal target check per completed game; many internal search steps may occur between them.

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Table 1 continued

Domain	Fast loop / proxy	Current AI cycle	Human / classical cycle	Target-level evidence τ_T	evi-	Limiting cycle	Approx. K / status
Low-level algorithm optimization	Correctness tests plus predicted or measured runtime.	AlphaDev trained for at most about 2 days in the reported tasks [M], explored at most 12 million programs in its worst reported case [M], and directly measured actual latency for less than 0.002% of generated programs [M] (Mankowitz, Michi, Zitnick, et al. 2023).	The optimized libc++ subroutines had not been changed for over a decade before the reported replacement [M] (Mankowitz, Michi, Zitnick, et al. 2023).	Correctness can be tested cheaply; true hardware latency is more expensive and therefore sampled sparsely.		Program execution on target hardware.	Very large between direct latency measurements; exact K depends on the training trajectory.
Chip floorplan-ning	Estimated wire-length, congestion, power/performance/area surrogates.	RL-generated floor-plans in under 6 hours [M] (Mirhoseini, Goldie, Yazgan, et al. 2021).	The source describes conventional floor-planning as requiring months of intense effort by physical-design engineers [M] (Mirhoseini, Goldie, Yazgan, et al. 2021).	Full downstream place-and-route, timing closure, physical verification, fabrication, and silicon performance provide progressively stronger checks.		EDA computation initially; ultimately fabrication and physical measurement.	[NQ]
Weather forecasting	Forecast generation and retrospective scoring against known historical weather.	GraphCast generates a 10-day global forecast in under 60 seconds [M] (Lam et al. 2023).	The reported HRES numerical forecast requires hours on a supercomputer for a comparable 10-day prediction [M] (Lam et al. 2023).	A 6-hour forecast can only be prospectively verified after 6 hours; a 10-day forecast after 10 days.		The physical evolution of the atmosphere.	Not an optimization K for a fixed forecaster; the generation/target ratio can nevertheless exceed 10^4 for a 10-day forecast.
General forecasting	Brier/log scores once questions resolve; before resolution, forecasters can update probabilities from new information.	ForecastBench starts new rounds every two weeks and samples 1,000 questions for LLMs and 200 for humans [M] (Karger et al. 2025).	Expert human forecasters are evaluated on the same future questions; the original study found experts outperforming the top tested LLM [M] (Karger et al. 2025).	Questions are explicitly about future events whose answers are unknown at submission; resolution is therefore event-limited.		Occurrence and reliable resolution of the event.	[NQ]; varies by question horizon.

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Table 1 continued

Domain	Fast loop / proxy	Current AI cycle	Human / classical cycle	Target-level evidence τ_T	evi- Limiting cycle	Approx. K / status
Autonomous materials synthesis	Synthesis outcome and automated characterization guide the next experiment.	A-Lab executed 353 experiments over 17 days [M], about 20.8 experiments/day platform throughput [D] (Szymanski et al. 2023).	The paper motivates automation partly by the slower conventional human-mediated workflow, but does not provide a clean matched per-experiment human cycle [NQ].	X-ray diffraction and other characterization can close the laboratory loop after synthesis; broader material performance may require additional testing.	Robotic handling, heating/reaction time, cooling and characterization.	Approximately one measured laboratory check per experiment for the measured property; do not infer serial latency from throughput.
Closed-loop chemistry	In-situ spectroscopy and reaction measurements feed subsequent experiment selection.	AlphaFlow used roughly 700 injection steps per campaign and produced more than 9,000 experimental conditions for learning [M] (Volk et al. 2023).	Conventional workflows require human transfer between design, execution, characterization and next-step selection; a matched wall-clock comparison is not reported [NQ].	The measured chemical state is available after the relevant reaction and characterization step.	Chemical kinetics, fluid handling and measurement.	Near one for the measured in-situ property; larger for unmeasured downstream properties.
Drug development and clinical medicine	Biomarkers, tumor response, progression-free survival and other surrogate endpoints can guide decisions earlier than patient-centered outcomes.	AI may accelerate candidate generation, but no general AI τ_P is defensibly comparable across drug programs [NQ].	Accelerated approval explicitly permits earlier decisions using surrogate endpoints (U.S. Food and Drug Administration 2024; Gyawali, Hey, and Kesselheim 2019).	Survival, function and quality of life may require years. In one 2009–2013 accelerated-approval cohort, half of required postapproval studies were completed only after at least 3 years [M] (Naci, Smalley, and Kesselheim 2017); later oncology analyses find unresolved clinical benefit beyond 5 years in some cases [M] (Liu, Kesselheim, and Cliff 2024).	Disease progression and sufficiently powered clinical follow-up.	Potentially very large, but the number of consequential decisions per target horizon is program-specific [NQ].

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Table 1 continued

Domain	Fast loop / proxy	Current AI cycle	Human / classical cycle	Target-level evidence τ_T	evi-	Limiting cycle	Approx. K / status
Reward-model alignment	A learned reward model, preference model, or direct preference loss is queried at every optimization step.	Optimization feedback is available every policy update; the useful independent axis is optimization budget or KL distance rather than wall-clock time [M/NQ] (Gao, Schulman, and Hilton 2022; Rafailov et al. 2024).	Human judgments are sufficiently costly that they are commonly distilled into a learned evaluator; no universal annotation latency exists [NQ].	Fresh human judgments or an independently held “gold” evaluator.		Human/evaluator throughput and coverage of the shifted policy distribution.	Directly variable experimentally via optimization strength.
Cybersecurity	Sandbox exploit success, tests, CTF flags, crash/reproduction signals.	CVE-Bench provides reproducible sandboxed real-world CVEs; the best reported agent framework exploited up to 13% of benchmark vulnerabilities [M] (Zhu et al. 2025).	Human security researchers also obtain fast reproduction feedback once a concrete vulnerability is isolated, but discovery and incident response times vary widely [NQ].	Real-world exploitability, prevalence, attacker adaptation and absence of undiscovered attack paths are not fully certified by sandbox success.		For a known exploit: [NQ] execution time. For “system is secure”: no finite exhaustive cycle in general.	
Recommendation / engagement optimization	Clicks, likes, watch time and other engagement responses are available almost immediately.	Recommendation systems can optimize real-time engagement [M] (Khambatta et al. 2023).	Human editors, advertisers and media organizations have analogous short-term audience metrics, but no matched cycle estimate is needed here [NQ].	User benefit or “time well spent” need not agree with engagement. In a preregistered experiment, actual-preference recommendations obtained more clicks (52% vs. 40%), while ideal-preference recommendations improved several measures of perceived user benefit [M] (Khambatta et al. 2023).		Depends on the chosen target; some welfare effects can be measured immediately, others only longitudinally.	Not necessarily large.
Public benchmarks and leaderboards	Benchmark score is available immediately after evaluation.	Models and prompts can be repeatedly selected against public or semi-public benchmark feedback.	Exams, scientific metrics and industrial KPIs provide analogous rapidly scored objectives.	External validity, contamination resistance and transfer to deployment may require held-out evaluation or later real-world evidence.		Independent held-out tasks or deployment.	Potentially large but usually not observable from public data.

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Table 1 continued

Domain	Fast loop / proxy	Current AI cycle	Human / classical cycle	Target-level evidence τ_T	evi-	Limiting cycle	Approx. K / status
Macroeconomic policy	Financial conditions, market rates, expectations and monthly/quarterly indicators provide relatively fast intermediate signals.	AI-assisted analysis can be generated much faster than macroeconomic effects unfold, but no defensible generic AI policy-update latency exists [NQ].	Central banks update policy and forecasts repeatedly while awaiting full transmission.	Federal Reserve summaries of the literature place maximal effects of a monetary-policy shock on activity and inflation at roughly 1–2 years [M] (Kugler 2025); the ECB likewise describes long, variable and uncertain lags (European Central Bank 2024).		Adjustment of contracts, borrowing, investment, wages, prices and expectations in the real economy.	[NQ]
Rare-event safety	Near misses, simulations, checklists, component tests and probabilistic risk models provide fast proxy evidence.	AI can run simulations and automated safety checks rapidly, but rare real failures remain event-limited [S/NQ].	A general-aviation accident investigation often requires 12–24 months to complete [M] (National Transportation Safety Board 2024); nuclear regulation explicitly supplements sparse operating experience with probabilistic risk assessment and precursor analysis (U.S. Nuclear Regulatory Commission 2024a; U.S. Nuclear Regulatory Commission 2024b).	The strongest evidence may require occurrence of a rare accident, followed by causal investigation.		Event frequency plus investigation latency.	Potentially enormous or undefined.
AI alignment and long-horizon safety	Behavioral evals, reward models, red-team suites, automated graders and held-out benchmarks.	Individual evals can be cheap and repeated, but no empirical literature currently supports a general target-evidence latency for robustness to future capability growth, strategic adaptation or deployment shift [NQ].	The closest human analogues are safety assurance, auditing, probabilistic risk assessment and incident investigation.	The relevant target may only be tested under new capabilities, distribution shift, prolonged deployment or strategic pressure.		Unknown; partly event- and capability-transition-limited.	Not yet empirically quantified.

6.1 Reading the Matched Horizons

Table 1 is easier to read once the domains are grouped by *why* their feedback geometry differs, not only by field name.

Matched verifiers. When the fast loop already checks the target criterion, a very large search or update count is compatible with safety. Formal theorem proving and rule-complete games are the clearest cases: Lean acceptance, terminal win/loss, and similar verifiers make proxy and target coincide, so effective $K \simeq 1$ even when internal search depth is enormous. The separate risk is semantic: a perfect checker on a mis-formalized statement is still the wrong target.

Mixed fidelity inside one artifact. Executable software often combines both regimes in a single codebase: $\tau_P \approx \tau_T$ for well-covered functional tests, but $\tau_P \ll \tau_T$ for security, maintainability, performance under real load, and rare bugs. Low-level algorithm optimization and chip floorplanning show a related *proxy hierarchy*: dense feedback from correctness checks, learned value estimates, or placement surrogates, while the stronger target—measured hardware latency, timing closure, silicon performance—is sampled only sparsely or arrives only downstream. AI can shorten construction time without making every downstream property available in the fast reward.

Clocks set by the world. Weather forecasting and general forecasting illustrate a different bottleneck. AI can compress candidate generation dramatically—GraphCast produces a ten-day forecast in under a minute—but prospective target evidence still waits on atmospheric evolution or event resolution. Neither humans nor models can manufacture the answer early; better reasoning can improve forecasts without closing the prospective target loop. This is where K can grow mechanically as optimization improves while τ_T stays fixed.

Shrinking τ_T through automation. Autonomous materials synthesis and closed-loop chemistry show the opposite response to physical delay. Self-driving laboratories do not merely generate hypotheses faster; they move characterization into the loop through robotics, in-situ measurement, and automated next-step selection. The limiting cycle becomes handling, reaction kinetics, and measurement rather than human handoff. Progress here attacks the target-evidence latency itself, not only τ_G or τ_P .

Institutional fast-proxy / slow-target structures. Clinical medicine is the strongest real-world example: surrogate endpoints can guide decisions years before survival, function, or quality of life are known, and regulators explicitly treat surrogates as predictors rather than direct measures of benefit. Macroeconomic policy provides a human analogue of acting repeatedly under delayed target effects—central banks revise policy on monthly or quarterly signals while full transmission to activity and inflation may take one to two years—but confounding shocks and a moving environment prevent a clean causal read on K .

Where the paper’s mechanism shows up directly. Reward-model alignment is the strongest direct AI evidence for accumulation under proxy selection: proxy scores can keep rising after independently evaluated quality has peaked and begun to decline. Public benchmarks add a different lesson: evaluator *independence* matters. A fast public score can lose informativeness once optimization is conditioned on its blind spots; dynamic or held-out evaluation is partly a response to that failure mode. Recommendation and engagement optimization are an important counterexample to a purely temporal theory: proxy failure can occur even when K is not large, because engagement and user benefit diverge immediately.

Local verifiers and open-ended targets. Cybersecurity combines unusually strong local feedback—sandbox reproduction, crash signals, CTF success—with a global target that has no finite exhaustive cycle in general. Demonstrating one exploit is much easier than certifying absence of exploitable paths. Rare-event safety is the limiting case of the horizon problem: because direct target observations are rare, mature fields construct surrogate feedback systems—probabilistic risk assessment, precursor analysis, investigation protocols—rather than waiting for repeated catastrophes.

Alignment as application, not evidence. The alignment row is the motivating application of the paper, not independent evidence for it. Reward-model overoptimization demonstrates the local mechanism; rare-event safety and clinical surrogate endpoints provide institutional analogues. Any stronger alignment extrapolation should remain explicitly conditional until target-evidence latency under capability growth, strategic adaptation, or deployment shift can be measured rather than assumed.

Two acceleration patterns. Two patterns recur across the table. First, AI can shrink both generation and target verification when the target is computationally checkable—formal proof, covered software properties, and terminal game outcomes. Second, AI can shrink generation or proxy evaluation while leaving target evidence nearly unchanged—weather, forecasting, clinical outcomes, and many deployment properties. The second regime is where faster optimization raises exposure unless independent target checks keep pace.

7 Feedback Timing in Human Learning

Educational feedback meta-analyses find medium average effects with large heterogeneity; information content strongly moderates benefit, and some feedback forms outperform others (Wisniewski, Zierer, and Hattie 2020). Meta-analyses of knowledge-of-results timing in motor learning find statistically significant timing effects with comparatively small effect sizes (Travlos and Pratt 1995). Delayed judgments can improve metacognitive accuracy by making long-term memory more diagnostic (Rhodes and Tauber 2011). Some test-learning experiments find delayed feedback beating immediate feedback (Butler, Karpicke, and Roediger 2007).

τ_P alone does not predict learning rate. A fast signal with little target information can underperform a slower, more diagnostic one. The human analogue is not “immediate feedback is good.” Learning tracks the information carried by feedback, and timing sets how much misleading optimization can run before better information arrives:

learning depends on the information carried by feedback,

with timing entering through the exposure window before correction.

8 Why AI Changes the Gap

AI changes the feedback geometry in a specific way. It can reduce τ_G , reduce some forms of τ_P , and increase parallel search capacity without changing the rate at which external reality produces independent evidence.

GraphCast is a clean illustration. A ten-day global forecast can be generated in under one minute, but the atmosphere cannot reveal the full ten-day error in under ten days (Lam et al. 2023). The compute-to-reality ratio has changed by orders of magnitude even though the physical target horizon has not.

Self-driving laboratories show a different response. Instead of merely accepting the physical delay, they attack it. A-Lab integrates planning, robotics, synthesis, characterization, and active learning in one loop, performing 353 experiments in 17 days (Szymanski et al. 2023). AlphaFlow uses in-situ measurements to turn intermediate physical states into feedback for subsequent experimental choices (Volk et al. 2023). These systems reduce the feedback horizon by building better measurement infrastructure around the physical process.

Clinical medicine compresses decisions onto surrogate endpoints because patient-centered endpoints take much longer to observe. Faster surrogate feedback pays off only when the surrogate tracks survival, function, or quality of life. Reviews of accelerated oncology approvals find many drugs approved on surrogate evidence that never show patient-centered gains, or remain uncertain years later (Gyawali, Hey, and Kesselheim 2019; Liu, Kesselheim, and Cliff 2024). Faster optimization has not shortened the biological target horizon; it has moved decisions onto an earlier, imperfect signal.

Behavioral evals, preference models, constitutional graders, and automated red teams supply rapid alignment signals. Robustness under capability growth, distribution shift, strategic behavior, or long deployment can still require target evidence on a much longer clock. A one-second score can proxy a one-year claim.

9 Verifier Construction: Moving the Target into the Fast Loop

So far we have largely treated the feedback structure as given. An optimizer receives a proxy every τ_P , while more independent target evidence arrives after τ_T . But this assumption hides one of the most important ways that difficult problems become tractable: the feedback architecture itself can be changed.

A useful intervention is therefore not only

generate better candidates,

but

make mistakes observable sooner and more faithfully.

We call this *verifier construction*. The term includes formal verifiers, tests, instruments, experiments, surrogate measurements, monitoring systems, and institutions that move target-relevant information into a shorter feedback loop.

9.1 Verification is constructed, not merely discovered

Many domains now treated as “objectively verifiable” became so only after substantial technical and institutional work. A physical quantity requires an operational measurement procedure. An experimental result requires controlled comparisons, calibration, and a way to distinguish signal from noise. A software property requires a specification and a test or proof obligation. A clinical claim requires an endpoint and a study design.

Thus the apparent binary distinction

verifiable vs. unverifiable

is often better represented as

weakly observable $\rightarrow$ instrumented $\rightarrow$ repeatably measurable $\rightarrow$ cheaply testable.

Scientific instruments are especially clear examples. An instrument can turn a latent state of the world into a repeatedly accessible signal. The important improvement is not necessarily that the world changes faster. Instead,

$$I(F; T)$$

increases, or informative F becomes available at shorter latency.

The same transformation occurs in engineering when an expensive full-system failure is replaced by component tests, simulation, telemetry, or formal invariants.

9.2 Verifier construction can reduce the feedback horizon in three ways

The framework separates three distinct improvements.

Earlier observation. A new measurement can reveal a relevant effect before the final outcome occurs. Continuous telemetry, biomarkers, in-situ spectroscopy, and precursor events are examples. If the earlier measurement is genuinely informative about the target, this reduces effective τ_T .

Cheaper repetition. Automation can make the same target-relevant experiment repeatable at much higher throughput. A-Lab integrates planning, synthesis, and characterization and executed 353 experiments in 17 days (Szymanski et al. 2023). AlphaFlow similarly couples experiment selection with in-situ measurement (Volk et al. 2023). Neither system removes chemical or physical latency, but both remove substantial human handoff latency and increase the number of target-relevant observations per unit time.

Decomposition. A long-horizon target can sometimes be split into shorter independently checkable conditions. Software tests provide a familiar example. A requirement such as “this service remains correct in deployment” is not directly verifiable in one short experiment, but type checks, unit tests, property tests, integration tests, security tests, and monitoring each constrain different failure modes.

In the ideal case,

$$T \approx g(T_1, \dots, T_m),$$

where each T_i has a much shorter verification horizon than T . The value of the decomposition depends on whether satisfying the pieces is sufficient for the whole. Missing interactions can recreate the original long-horizon problem.

9.3 Surrogates are compressed verifiers

A common way to shorten the feedback horizon is to replace a slow target T with an earlier observable P :

$$T \longrightarrow P \longrightarrow \text{fast optimization.}$$

This is verifier construction too, but it creates the central risk studied in this paper. A surrogate trades latency for approximation error.

Clinical medicine makes the trade explicit. Accelerated approval can rely on surrogate endpoints that are expected to predict clinical benefit, allowing decisions before survival or quality-of-life evidence fully matures (Gyawali, Hey, and Kesselheim 2019; Naci, Smalley, and Kesselheim 2017). Later studies sometimes fail to show corresponding patient-centered benefits (Liu, Kesselheim, and Cliff 2024). The surrogate has successfully shortened the decision loop while imperfectly shortening the *target* loop.

Reward models are the analogous construction in preference optimization. Expensive human judgments are compressed into a model that can be queried at machine speed. This creates enormous evaluation throughput but also creates an exploitable residual: continued optimization can increase proxy reward while independent evaluation degrades (Gao, Schulman, and Hilton 2022; Rafailov et al. 2024).

Verifier construction therefore creates a trade:

$$\text{latency reduction} \quad \text{versus} \quad \text{loss of target information.}$$

9.4 A verifier is useful only relative to a claim

There is no universal verifier for an artifact.

Lean verifies formal validity, not that the theorem is the one the human intended. A unit test verifies tested behavior, not global program correctness. A vulnerability reproduction harness verifies the presence of one exploit, not the absence of others. A biomarker measures a biological response, not necessarily patient benefit.

We can express this as a chain:

$\text{target specification} \rightarrow \text{observable} \rightarrow \text{verifier} \rightarrow \text{optimization.}$

Failure can occur at each arrow.

This also separates *certification* from *construction*. Given a candidate x , an inexpensive verifier may answer whether x satisfies a criterion without providing an efficient method for finding such an x . Formal theorem proving makes the distinction especially clear: proof checking is cheap compared with proof search. Verifier construction improves the selection landscape; it does not by itself solve the search problem.

9.5 Optimization changes the requirements on the verifier

A verifier that works for randomly sampled candidates need not remain reliable on candidates selected specifically to maximize it. Let $P_0(x)$ be the distribution on which an evaluator was validated and $P_k(x)$ the distribution after k rounds of optimization. What matters is not only

$$\text{Corr}_{P_0}(F, T),$$

but whether the relationship survives the shift

$$P_0 \rightarrow P_1 \rightarrow \dots \rightarrow P_K.$$

Reward-model overoptimization provides direct evidence of this problem (Gao, Schulman, and Hilton 2022). Continued policy optimization moves samples into regions where the learned evaluator is less reliable. More generally, optimization searches for the residual error of the verifier.

A useful verifier for a powerful optimizer must therefore satisfy a stronger criterion than ordinary predictive accuracy:

high fidelity under the distribution induced by optimizing the verifier.

This requirement becomes especially difficult when the optimizer can influence the measurement process itself.

9.6 Independent correction is part of verifier design

Verifier construction should therefore not aim only at making a single evaluator faster. It can also construct *independent* correction channels.

Suppose a fast evaluator F_1 guides most optimization, while evaluators $F_2, \dots, F_m$ are queried less frequently. The secondary evaluators are valuable when their errors are not strongly correlated:

$$\text{Cov}(\epsilon_i, \epsilon_j) \text{ is small,} \quad \epsilon_i = F_i - T.$$

A second evaluator produced from the same data, architecture, and objective may add little independent information even if its outputs look different. Independence is therefore a causal property of the evaluation process, not merely an ensemble size.

Human institutions often approximate this architecture through replication, independent audits, adversarial review, appeals, and multiple measurement methods. These mechanisms are imperfect, but their structural purpose is relevant here: they prevent the process being optimized from relying indefinitely on one unchecked error channel.

9.7 Verifier construction is the main escape from a growing horizon gap

If AI reduces τ_G and τ_P while τ_T remains fixed, then K grows. One response is to slow optimization. A more powerful response, when possible, is to redesign the evidence channel:

$$\begin{array}{c} \tau_G \downarrow \quad \tau_P \downarrow \quad \tau_T \text{ fixed} \\ \Downarrow \\ K \uparrow \end{array}$$

versus

$$\begin{array}{c} \tau_G \downarrow \quad \tau_P \downarrow \quad \tau_T \downarrow \\ \Downarrow \\ K \text{ controlled.} \end{array}$$

This gives a different interpretation of scientific and engineering progress. Capability improvements search the solution space faster. Measurement, instrumentation, testing, and experimental design make the consequences of that search legible faster.

The two are complements. Fast search with a strong verifier produces the favorable regime represented by theorem proving and well-tested software. Fast search with a weak but rapidly computed proxy produces the regime represented by reward-model overoptimization.

For AI safety, the practical question is therefore not merely whether an evaluation suite exists. It is:

Does the evaluation architecture make target-relevant errors visible at roughly the rate at which optimization can discover and amplify them?

If not, increasing evaluator throughput alone may make the proxy loop faster without making the correction loop faster. The feedback horizon has not actually been closed.

10 Scope and Limits

10.1 Perfect or sufficient proxies

When proxy feedback matches the target, a long horizon gap costs nothing. Formal theorem checking on a fixed statement is the standard example.

10.2 Search difficulty

Formal proofs and NP-hard search problems can be cheap to verify and expensive to find. Verification latency and search hardness are separate axes.

10.3 Plural targets

For social welfare, governance, or alignment, there may be no single scalar L_T . “Target evidence” then means an independent check on whichever outcome dimension matters for the decision; K depends on that choice.

10.4 Parallelism

The ratio τ_T/τ_P understates exposure when many candidates run in parallel and overstates it when updates are tiny or reversible. Count consequential proxy-guided updates between target checks directly.

10.5 The AI–human analogy is structurally asymmetric

The matched examples in this paper should not be read as claiming that AI optimization is simply accelerated human problem solving. The comparison is useful because the same feedback structures recur in both cases, but several important mechanisms are asymmetric.

AI has much greater parallel search. A human expert normally explores a small number of candidate solutions at once. Digital systems can generate, execute, and reject enormous candidate sets in parallel. AlphaProof trains on millions of formally checked problems (Hubert et al. 2025); AlphaDev explored millions of candidate programs (Mankowitz, Michi, Zitnick, et al. 2023). Consequently, wall-clock ratios such as τ_T/τ_P can substantially understate the amount of selection occurring between target checks. This is one reason we define K primarily as the number of consequential proxy-guided updates, rather than as a time ratio.

AI state can be copied and branched cheaply. Human learning is transmitted imperfectly through teaching, writing, imitation, and institutions. A trained digital state can instead be copied, checkpointed, branched, evaluated, and recombined with comparatively little loss. A useful update can therefore propagate across many simultaneous search processes without requiring each process to rediscover it. There is no close human analogue at the level of individual cognition.

AI can be optimized internally. Human feedback primarily selects actions, strategies, institutions, or people. Machine learning also changes the internal parameters that generate subsequent behavior. Gradient-based optimization can therefore perform many small internal updates between externally meaningful decisions. Whether each gradient step should count toward K depends on the analysis. For safety questions, the relevant unit is normally the smallest update capable of producing a consequential change in the system’s policy or deployment behavior, not every floating-point parameter update.

Human progress has stronger examples of verifier invention. The asymmetry also runs in the other direction. Human scientific progress repeatedly changed not only candidate solutions but the *measurement regime*: new instruments, standardized measurements, controlled experiments, statistical methods, and institutions made previously weakly observable claims testable. Current AI systems can propose experiments and help design measurement procedures, and self-driving laboratories automate existing

instances (Volk et al. 2023; Szymanski et al. 2023). But the historical record contains much stronger examples of humans changing what could be observed at all.

Humans can sometimes learn from $N = 1$ events. Wars, accidents, financial crises, constitutional failures, and scientific anomalies may occur too rarely for ordinary repeated optimization. Humans nevertheless build causal narratives, compare mechanisms across cases, construct counterfactuals, and change institutions after unique events. These updates can be wrong, but they are not well represented as repeated trials from a stationary distribution. Present AI training, by contrast, derives much of its strength from large repeated datasets or synthetically generated task families.

Human institutions can deliberately separate evaluators. Science, auditing, courts, safety engineering, and other institutions often create partially independent channels of criticism. Their purpose is not merely to increase feedback frequency but to prevent one selection criterion from controlling all evidence about itself. AI evaluation often compresses many judgments into a single benchmark or reward model. Multiple AI critics do not automatically recreate institutional independence if they share training data, architecture, prompts, or failure modes.

The comparison is therefore conditional. The paper makes a comparison at the level of feedback architecture:

candidate $\rightarrow$ selection signal $\rightarrow$ update $\rightarrow$ later target evidence.

It does not claim equivalence between human cognition, scientific institutions, and machine learning. The important empirical question is narrower: holding the target and evaluator structure approximately fixed, what changes when the number of proxy-guided selections between independent target checks increases?

This scope restriction matters for interpreting the historical examples. The human record motivates the variables and provides useful analogies, but it does not by itself establish a causal law for AI systems. Conversely, reward-model overoptimization provides direct evidence for the accumulation mechanism in AI but does not establish that institutional or scientific systems obey the same quantitative relationship.

11 Discussion

The operative distinction is not easy versus hard problems. It is whether optimization runs inside a loop whose evaluator is both fast and tied to the target.

That split explains rapid AI progress in formal mathematics, games, and test-covered software. Candidate generation is cheap, and these domains already ship reliable selectors. AlphaProof runs large-scale RL because Lean supplies grounded proof feedback (Hubert et al. 2025). Code models self-debug because execution makes some failures legible (Chen et al. 2023). The verifier need not know how the candidate was produced.

The same structure inverts when the evaluator is imperfect. Gao et al. show proxy reward rising while an independent evaluation falls under continued optimization (Gao, Schulman, and Hilton 2022). Per-update target errors sum for as many updates as the system takes before correction. Faster optimization raises that count unless independent evaluation keeps pace.

The local question behind “capabilities may outrun alignment” is concrete:

How many consequential optimization updates can occur before an evaluator with sufficiently independent failure modes gets to reject the direction of travel?

Sometimes the answer is one: a failed Lean proof rejects immediately. Sometimes it is millions: a fixed reward model guides a long RL run before fresh human preference data arrives. Sometimes physics sets the clock: a ten-day forecast needs a ten-day world. Sometimes biology or institutions do: a surrogate endpoint can precede patient-centered clinical evidence by years.

Evaluation research shortens the target-relevant horizon or it does not. A faster proxy score without faster independent target evidence increases optimization speed without increasing correction speed.

12 Conclusion

The Feedback Horizon Gap is the count

$$K = \text{proxy-guided optimization updates between independent target checks.}$$

For regular sequential systems, $K \approx \tau_T/\tau_P$. Target drift over that window is

$$L_T(\theta_K) - L_T(\theta_0) = \sum_i \Delta_i.$$

Target-safe proxy updates make large K cheap. Systematic or correlated proxy error accumulates until independent evidence arrives.

Reward-model overoptimization shows the failure mode in AI. Formal theorem proving and execution-tested software show large throughput with strong verifiers. Weather forecasting, autonomous laboratories, and clinical surrogate endpoints show optimization and target evidence on mismatched clocks.

As optimization accelerates, reliability tracks how many consequential updates get selected before sufficiently independent target evidence arrives, not how quickly a proxy returns a score. That count is readable in many AI systems today; cross-domain prediction beyond the reward-overoptimization cases remains to be tested.

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